Explicit–Implicit Semantic Collaborative Guidance for Cross-Modal Remote Sensing Image–Text Retrieval

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***Abstract*—Cross-modal remote sensing image–text retrieval (CMRSITR) is an approach that integrates remote sensing images with textual descriptions, leveraging the semantic correlation between images and text to perform efficient and accurate information retrieval. Existing methods typically rely entirely on implicit semantic alignment, leading to issues with semantic gaps and insufficient fine-grained retrieval capability. In this article, we propose an explicit–implicit semantic collaborative guidance (EISCG) framework, which enhances retrieval accuracy and efficiency by constructing parallel explicit and implicit semantic branches. The implicit branch (IB) obtains robust global semantic representations by aligning the global feature vectors of images and text. The explicit branch (EB) introduces a context-aware TF-IDF (CT) weighting strategy to dynamically assess the importance of words within specific contexts, achieving fine-grained word-to-region supervision and providing precise local supervision signals to the IB. Furthermore, we design a differentiable adaptive binarization projection (DABP) module, integrated at the end of the IB, which can transform the generated high-dimensional characteristics into sparse binary hash codes, achieving millisecond-level retrieval while maintaining high accuracy. Experiments across a variety of datasets for RSITR demonstrate that EISCG exceeds current state-of-the-art methods in both recall and retrieval speed, thus demonstrating strong robustness and scalability.**

***Index Terms*—Cross-modal remote sensing image–text retrieval (CMRSITR), explicit branch (EB), explicit–implicit semantic collaborative guidance (EISCG), implicit branch (IB).**

I. Introduction

With the advancement of satellite technology, types and scale of RS data [1] are experiencing exponential growth. High-resolution imagery and large-scale geographic data provided by various remote sensing satellites have created an unprecedented wealth of Earth observation resources [2], [3], [4]. However, efficiently and accurately extracting useful information from these massive amounts of imagery has become a critical challenge in remote sensing. Since RS images inherently include abundant spatial information and geographic details, image-based retrieval methods [5], [6], [7] often face challenges such as high dimensionality, semantic ambiguity, and scene diversity. Meanwhile, remote sensing data are often accompanied by rich metadata, descriptive text, and manual annotations. Joint retrieval of image–text data helps to address the limitations of single-modal retrieval and enhances the effectiveness of information utilization. Therefore, to better resolve the semantic gap between remote sensing image–text information, CMRSITR [8], [9] has emerged and is gradually employed in fields such as disaster monitoring, environmental protection, and resource management [10], [11], [12], [13]. It is anticipated to emerge as a key tool for future intelligent interpretation of RS data.

Despite its significant application potential, CMRSITR faces core challenges in its development. The differences between image and text modalities lead to difficulties in semantic alignment, which in turn affects retrieval accuracy. Effectively bridging the semantic gap and achieving robust cross-modal matching remains the focus of research in this field.

Current RSITR methods often follow an implicit semantic alignment paradigm, where global feature vectors of texts and images are mapped to a shared semantic space. These methods can learn robust global representations with good generalization ability. However, they still face limitations in capturing fine-grained semantic correspondences [14], [15], [16], [17], resulting in a persistent semantic gap. In recent years, some studies have attempted to introduce explicit alignment mechanisms to enhance local semantic relationships. However, these approaches typically treat explicit alignment as an auxiliary module rather than an independent and systematic supervision branch, limiting their improvement. Therefore, balancing explicit and implicit semantic information while maintaining generalization ability and improving fine-grained alignment quality remains a key challenge in current research.

To address the semantic gap and the limited fine-grained retrieval capability caused by the conventional single implicit alignment paradigm, we propose an explicit–implicit semantic collaborative guidance (EISCG) framework for accurate and efficient RSITR, as shown in Fig. 1.

The proposed framework is composed of two parallel branches: an explicit semantic guidance branch and an implicit semantic guidance branch. The explicit branch (EB) estab-

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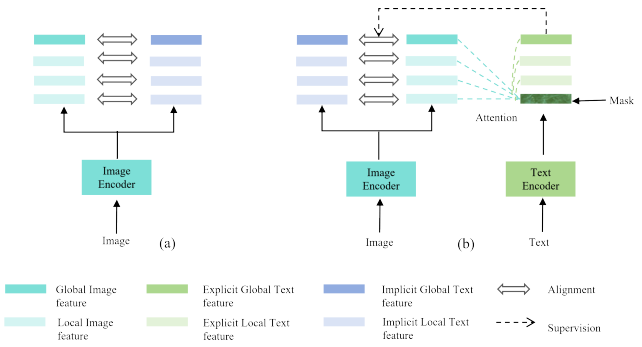


Fig. 1. Comparison of the existing CMRSITR methods with our proposed method. (a) Typical implicit semantic guidance method, while (b) EISCG framework designed by us.

lishes fine-grained correspondences between words and image regions, providing precise local supervision signals to guide representation learning, while the implicit branch (IB) focuses on learning global cross-modal representations and performs the final retrieval matching. Through the collaborative interaction of the two branches, EISCG effectively combines the strong generalization ability of implicit alignment with the high precision of explicit alignment, thereby significantly enhancing retrieval performance in complex remote sensing scenarios. To further enhance the fine-grained supervision capability of the EB, we design a context-aware TF-IDF weighting strategy. Unlike conventional term-frequency statistics [18], our strategy incorporates sentence-level semantic information to dynamically assess word importance under a given context. By leveraging semantic embeddings to compute contextual similarity and integrating an attention mechanism, this strategy can accurately identify key semantic units in the text and selectively mask less informative tokens. This design not only strengthens the model's ability to capture deep textual semantics but also provides more precise and efficient local supervision for cross-modal alignment, thereby effectively improving semantic association performance in complex remote sensing scenarios.

In addition, to mitigate the increased computational overhead introduced by the explicit–implicit dual-branch architecture, we propose a lightweight differentiable adaptive binarization projection (DABP), which is integrated at the end of the IB to map high-dimensional and dense cross-modal fused features into compact binary hash codes. By introducing a differentiable binarization function together with a multinomial-distribution-based balance regularization constraint, DABP enables end-to-end optimization while ensuring both uniform distribution and strong discriminability of the generated hash codes in the Hamming space. This design significantly improves retrieval efficiency with negligible additional computational cost, making the proposed framework well suited for real-time retrieval in large-scale remote sensing image archives. Through these innovations, the EISGF framework not only effectively alleviates the semantic gap inherent in conventional implicit paradigms and substantially enhances fine-grained retrieval performance but also maintains high retrieval efficiency, providing a novel and practical solution for RSITR.

The contributions of our work are as follows.

1) We propose a novel EISCG framework, which effectively integrates the fine-grained alignment capability of explicit methods with the strong generalization ability of implicit methods.
2) We design a context-aware TF-IDF strategy, enabling the precise identification of key semantic units and selective masking enhancement, thereby providing more accurate and efficient local supervision signals for cross-modal alignment.
3) We propose the DABP module, which efficiently maps high-dimensional fused features into compact binary hash codes, significantly improving retrieval efficiency while maintaining almost no loss in accuracy.
4) We conduct extensive experiments on multiple publicly available benchmark datasets. The experimental results show that our method achieves an excellent balance between retrieval accuracy and efficiency on datasets like RSICD [19], RSITMD [20], and UCM [21], fully validating its robustness, effectiveness, and scalability.

The organization of this article is as follows. Section II provides an overview of related work, with a focus on advancements in CMRSITR and VLP models. Section III presents a detailed description of the proposed EISCG framework, along with the design principles behind its core components. Section IV validates the effectiveness of the method through comprehensive comparative and ablation experiments. Finally, Section V concludes the article by summarizing the key findings and contributions and outlining potential directions for future research.

II. Related Work

A. Cross-Modal RS Image–Text Retrieval

CMRSITR is a vital branch of multimodal learning, with the core objective of establishing semantic associations between the two heterogeneous modalities, image and text, facilitating image-to-text (I2T) or text-to-image (T2I) search. Current approaches can be divided into title-based and embedding-based techniques.

Title-based generation methods aim to generate descriptive text for images, which is then used for retrieval through text matching. These methods typically utilize keywords or structured information in the title to enhance the association with visual content. Chen et al. [22] introduced the title-guided network, a model that treats the title as a class query input, highlighting its pivotal role in the document's overall structure. Mishra et al. [23] introduced a new method that uses a pretrained Transformer model, GPT-2, to automatically generate titles for a given text. This method first generates some titles, then selects suitable ones from it for denoising and refinement operations, ultimately yielding the final title. Amirian et al. [24] developed a system that combines image and video description methods with natural language processing tools to generate short summaries and titles for videos. Yang et al. [25] have presented a methodology for the

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automatic generation of titles for new products with limited labels. This methodology is based on a prompt-based method and a multimodal prompt learning framework. To fully explore the visual associations in images, Yuan et al. [26] proposed a multilayer attention-based architecture for generating subtitles for RS images. In addition, Ma et al. [27] designed two multiscale approaches to generate suitable captions, with both methods combining different features to enhance each other. Furthermore, Ye et al. [28] proposed a two-stage joint training RSI captioning method. This method addresses the issues of long processing times when image description and auxiliary remote sensing tasks are handled separately, as well as the interference between tasks. Although title-based methods can quickly match relevant information, the titles may suffer from ambiguity or unclear expression, which can lead to a reduction in retrieval quality.

In the context of recent advances in deep learning, embedding-based retrieval methods have become mainstream. These methods achieve efficient cross-modal retrieval by mapping images and texts into a shared semantic space and calculating similarities. Cheng et al. [29] introduced a new network and devised a semantic alignment model to facilitate the exploration of potential correspondences between texts and images. In addition, to surmount the constraints of intrinsic relationships between different features, Wang et al. [30] emphasized the inherent relationships between different features, starting with simple features from training samples to achieve model lightweighting. Yuan et al. [20] extracted image features through a multiscale attention mechanism and used these features to guide text representation. A fine-grained dataset was simultaneously constructed to support keyword-based image retrieval. Furthermore, Yuan et al. [31] designed a model capable of filtering redundant features, thereby significantly reducing computational complexity while implementing supervised optimization to ensure model accuracy. Then, Zhao et al. [32] introduced a framework called masked interactive inference and alignment (MIIA) utilizing dynamic contrastive learning (DCL) to achieve the critical alignments required for effective cross-modal matching. Subsequently, Zhao et al. [33] proposed a sparse-guided partial dense (SGPD) cross-modal retrieval method, which performs sparse mapping of the dense search results through a sparse projection head. It leverages overlaps between sparse retrieval results and dense vector clusters to achieve high-precision and fast retrieval. Finally, to strike a balance between efficiency and accuracy, Zhou et al. [34] proposed a novel CFITR framework, which integrates dual-stream and single-stream architectures into a two-stage retrieval process.

Beyond the aforementioned embedding-based retrieval methods, recent advances in multimodal learning and fine-grained recognition offer complementary perspectives that are highly relevant to cross-modal remote sensing image–text retrieval (CMRSITR). Tang et al. [35] propose the CONTRINET, a robust confluent triple-flow network employing a "Divide-and-Conquer" strategy. Tang et al. [36] propose an effective bidirectional pyramid architecture to enhance internal representations of features. Tang et al. [37] propose a matching-based framework named M3Net for few-shot fine-grained action recognition. In [38], SYNTRANS transfers diverse knowledge from large multimodal models to boost few-shot learning. Tang et al. [39] introduce CoEvo, a training-and annotation-free test-time framework that performs bidirectional, sample-conditioned adaptation of both textual and visual proxies.

B. Vision–Language Pretraining Model

VLP learns joint representations of vision and language by pretraining on image–text pairs [40]. Driven by the advancements in NLP pretraining language models (such as BERT [41]/GPT [42]), VLP has gradually become the core training paradigm for current VL tasks and is widely applied to downstream tasks like image captioning, visual question answering, and image–text retrieval.

From an architectural design perspective, existing VLP methodologies are generally categorizable into two types: single-stream and multistream models. Single-stream models (UNITER [43], ViLT [44]) typically map visual and linguistic representations into a unified sequence space and jointly model them within a shared Transformer to explicitly capture cross-modal interactions. However, such methods often require self-attention computations on long concatenated sequences, leading to high computational and storage costs and offer limited flexibility in independently optimizing intramodal representations. On the contrary, multistream models (ViLBERT [45], ALBEF [46]) decouple the visual and linguistic streams through separate encodings, with information fusion achieved through a cross-modal interaction module at intermediate or higher layers, thereby enhancing cross-modal alignment and understanding while preserving intramodal representational capacity. Among them, CLIP [47] is a groundbreaking work that employs an extremely simple dual tower architecture and a pure image–text contrastive learning objective. The model has been trained on an extensive dataset comprising 400 million image–text pairs, thereby demonstrating notable zero-shot transfer efficacy. Consequently, it can be directly applied to tasks like image classification and image–text retrieval without fine-tuning. In addition, CLIP's publicly available models and code have become the foundation for subsequent VLP and LLMs, such as BLIP [48] and LLaVA [49].

In recent years, pretraining models have become the foundational models in remote sensing image research. By leveraging transfer learning, these models inject knowledge learned from large-scale data into specific tasks, significantly improving analysis performance and application efficiency. Xiu et al. [50] used a CLIP-based model that was continuously pretrained on an RS dataset of five million pairs, focusing on cross-modal image–text retrieval tasks. They also introduced an efficient framework, RSITR-FFT, which refines the feature space by incorporating fine-grained word-region alignment and integrating consistency constraint regularization terms into the learning objective. In addition, to effectively utilize local information, Chen et al. [51] introduced a global and local image captioning (GLCap) method. By integrating global and local captions through large language models (LLMs), they achieved higher quality captions, addressing the scarcity of high-quality image–text datasets during fine-tuning RS VLM.

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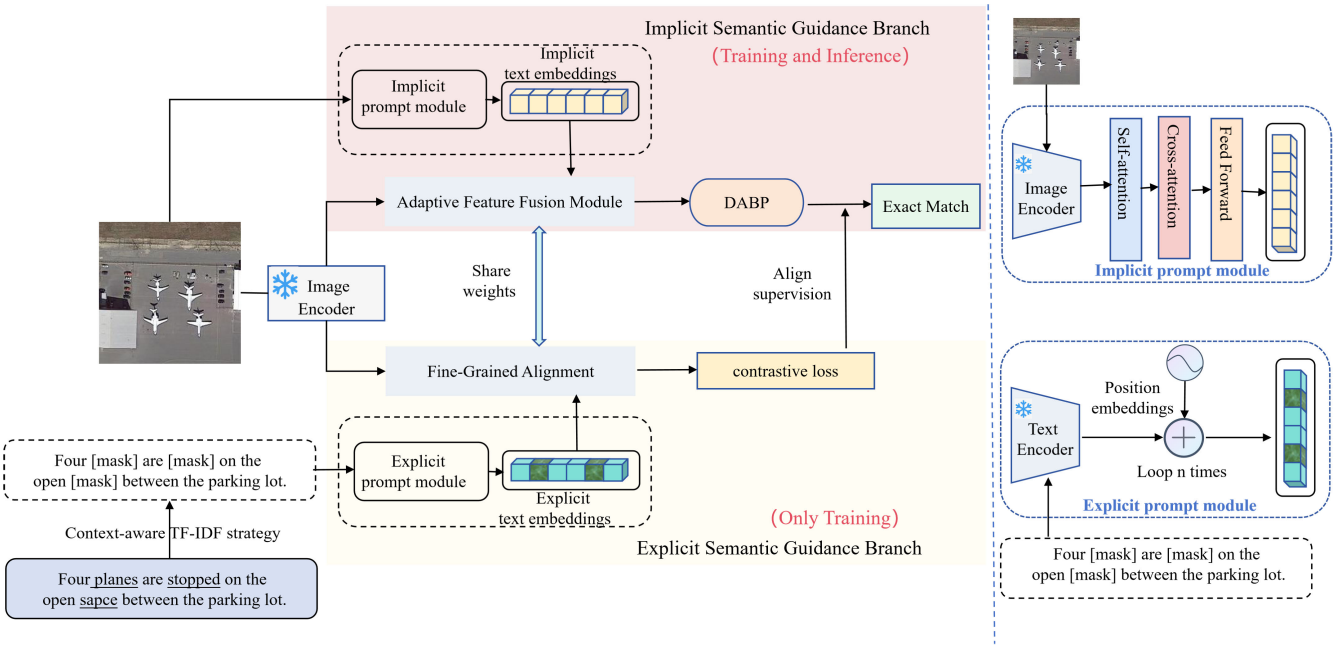


Fig. 2. Overview of the EISCG architecture. The framework consists of an explicit semantic guidance branch, an implicit semantic guidance branch, a DABP head, and a CT strategy. The EB establishes fine-grained correspondences between words and image regions through contrastive learning, providing precise local supervision signals for the IB. On the right are the implicit prompt module and the explicit prompt module.

Based on the above, existing methods either rely solely on global implicit alignment, leading to the loss of fine-grained information, or treat explicit alignment as an auxiliary module, failing to fully exploit its potential. To address these issues, this article proposes the EISCG framework, which treats explicit alignment as an independent branch working in parallel with the IB to achieve deep integration of global and local semantics. At the same time, although pretrained models provide powerful initialization for cross-modal tasks, targeted fine-tuning and structural design are still required in the remote sensing domain.

III. Method

This section provides a detailed description of the proposed EISCG framework. As shown in Fig. 2, the framework consists of three core modules: the explicit semantic guidance branch (Section III-B), the implicit semantic guidance branch (Section III-C), and the DABP module (Section III-D). The EB establishes fine-grained word-region correspondences using optimal transport (OT), providing local supervision signals to the IB; the IB learns global representations and is responsible for final retrieval; the DABP module converts the output of the IB into compact hash codes to improve efficiency. The three modules are jointly optimized through multitask learning (Section III-E).

A. Feature Extraction Backbone Network

Remote sensing images possess distinct characteristics, such as large scale, multispectral bands, and complex distribution of ground objects, which differentiate them from natural images. To fully leverage large-scale visual-linguistic prior knowledge, this framework employs a CLIP model pretrained on massive natural image–text pairs as its backbone network. Its strong generalization ability provides a solid foundation for cross-modal understanding of remote sensing data.

Given a training dataset $\mathcal{D} = \{(I_i, T_i)\}_{i=1}^{N}$, which contains N image–text pairs, where I_n is the nth remote sensing image and T_n is its corresponding text description. We use the ViT-B/16 architecture as a dual-stream encoder. The visual encoder $f_v(\cdot)$ is based on the Vision Transformer architecture. The input image I_n is divided into $P = (H \times W)/16^2$ nonoverlapping image patches and projected linearly to obtain a sequence of patch embeddings, with an additional learnable [CLS] token. After processing through Transformer layers, the output corresponding to the final [CLS] token serves as the global image feature vector $\mathbf{v}_n = f_v(I_n) \in \mathbb{R}^{d_v}$.

In addition, we extract all patch feature sequences from the last Transformer layer, $\mathbf{V}_n = \{\mathbf{v}_n^1, \mathbf{v}_n^2, \ldots, \mathbf{v}_n^P\} \in \mathbb{R}^{P \times d_v}$, which retain the local spatial information of the image, providing the basis for fine-grained alignment.

The text encoder $f_t(\cdot)$ is also based on the Transformer architecture. The input text T_n is tokenized, and special tokens are added before being passed through the text encoder to obtain the embedded sequence of token representations. Similarly, the output corresponding to the [EOS] token serves as the global text feature vector $\mathbf{t}_n = f_t(T_n) \in \mathbb{R}^{d_t}$. We also extract all token feature sequences $\mathbf{T}_n = \{\mathbf{t}_n^1, \mathbf{t}_n^2, \ldots, \mathbf{t}_n^L\} \in \mathbb{R}^{L \times d_t}$, where L is the sequence length.

During training, to stabilize the process and retain the pretrained knowledge, the parameters of the backbone network are kept frozen. Only the parameters of the subsequently introduced modules are optimized.

B. Explicit Semantic Guidance Branch

1) Context-Aware Dynamic Keyword Extraction: Traditional TF-IDF measures the importance of words solely from

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a statistical perspective, without capturing the semantic value of words in specific contexts. For example, in the description of dense buildings in the city center, the TF-IDF value of the word dense may not be high, yet its modifying role is crucial for understanding the scene. To address this, we propose a context-aware dynamic keyword extraction strategy.

We first compute a corpus-level TF-IDF score as a statistical prior for each token. For a token w_j in a sentence T_n, its TF-IDF score is defined as follows:

$$w_j^{tfidf} = tf\left(w_j, T_n\right) \cdot idf\left(w_j\right) \tag{1}$$

where $tf(w_j, T_n)$ denotes the term frequency of token w_j in sentence T_n. The inverse document frequency is computed over the entire training corpus

$$idf\left(w_j\right) = \log \frac{N_s}{1 + df\left(w_j\right)} \tag{2}$$

where N_s denotes the total number of sentences in the training corpus and $df(w_j)$ represents the number of sentences containing token w_j.

It should be emphasized that the TF-IDF scores are precomputed at the corpus level before training and remain fixed during optimization. They are not computed within each minibatch, thus avoiding unstable batch-level IDF estimation.

Although the TF-IDF prior itself is static, the final word importance is dynamically adjusted according to the sentence context. We use a lightweight bidirectional Transformer encoder to deeply encode the text token features T_n, obtaining context-enhanced tokens $\mathbf{T}_n^{\text{ctx}} = \{\mathbf{t}_n^{\text{ctx},1}, \ldots, \mathbf{t}_n^{\text{ctx},L}\}$. This encoder captures long-range dependencies between words. By performing average pooling on $\mathbf{T}_n^{\text{ctx}}$, we obtain the global context vector c_n. The cosine similarity between each context-enhanced token $\mathbf{t}_n^{\text{ctx},j}$ and the global context c_n is computed as its semantic relevance score s_j^{sem}

$$s_j^{sem} = \cos\left(t_n^{ctx,j}, c_n\right). \tag{3}$$

Finally, the dynamic importance weight w_j^{dynamic} of each token is determined by both its TF-IDF weight w_j^{tfidf} and its semantic relevance score

$$w_j^{\text{dynamic}} = \frac{\exp\left(\beta \cdot w_j^{\text{tfidf}} + (1-\beta) \cdot s_j^{\text{sem}}\right)}{\sum_{k=1}^{L} \exp\left(\beta \cdot w_k^{\text{tfidf}} + (1-\beta) \cdot s_k^{\text{sem}}\right)} \tag{4}$$

where $\beta \in [0, 1]$ is a learnable parameter used to enable the model to adaptively balance statistical and semantic information.

During training, the precomputed TF-IDF scores are treated as fixed priors, while the context encoder and the balancing coefficient β are optimized through the downstream OT alignment loss $\mathcal{L}_{\text{align}}$ (via backpropagation). This module serves as the preprocessing stage for the EB, and its parameters are updated end-to-end through the same loss, without an independent loss function. While optimizing the transport plan I^*, the alignment loss implicitly guides the learning of the dynamic importance weights w_j^{dynamic}, enabling the model to automatically identify and strengthen the key semantic units that contribute most to image–text matching. In this way, the proposed weighting strategy preserves the robustness of corpus-level lexical statistics, allows the model to adaptively emphasize contextually important semantic units, and avoids complex auxiliary supervision signals, ensuring the simplicity and efficiency of the entire EB training.

2) Fine-Grained Alignment Based on OT: To establish word-region correspondences, we model it as an OT problem, aiming to transport the text semantics to the image regions with minimal cost. OT theory enables the establishment of local geospatial-region correspondence. Compared to traditional methods (which focus solely on quantity-based regions), OT offers broader global coverage, avoiding information loss. In the mathematical framework, we distribute image blocks evenly for the OT of visual data to text-based dynamic distributions, seeking the OT plan Γ^* and constructing the alignment loss L_{align}, which guides the model to learn precise local correspondences.

We compute the cosine similarity matrix $S \in \mathbb{R}^{P \times L}$ between the image patch features $\mathbf{V}_n$ and the context-enhanced text features $\mathbf{T}_n^{\text{ctx}}$, and use its negative values as the transport cost matrix $C = 1 - S$. The constant term in $C = 1 - S$ denotes an all-one matrix $\mathbf{1}_{P \times L}$ with the same size as S. Therefore, the transport cost matrix is computed element-wise as $C_{ij} = 1 - S_{ij}$, where S_{ij} is the cosine similarity between the ith image patch feature and the jth text token feature. A higher semantic similarity leads to a lower transport cost, encouraging semantically matched word-region pairs to receive larger transport mass. The Frobenius inner product is defined as $\langle \Gamma, C \rangle_F = \sum_{i=1}^{P} \sum_{j=1}^{L} \Gamma_{ij} C_{ij}$, which measures the total transport cost under the transport plan Γ. We solve the following entropy-regularized OT problem to obtain the transport plan Γ^*:

$$\Gamma^* = \arg \min_{\Gamma \in \Pi(a,b)} \langle \Gamma, C \rangle_F + \epsilon H\left(\Gamma\right) \tag{5}$$

where $\Pi(a, b) = \{\Gamma \in \mathbb{R}_+^{P \times L} \mid \Gamma 1 = a, \Gamma^T 1 = b\}$ represents the feasible domain of the transport plan. We set the marginal distribution $a = 1/P 1^P$ representing the uniform distribution across all image blocks, and the marginal distribution of the text words $b = w^{\text{dynamic}}$, which is the dynamic word distribution obtained from TF-IDF weighting. The entropy regularization term is $H(\Gamma) = -\sum_{i,j} \Gamma_{i,j}(\log \Gamma_{i,j} - 1)$, where ϵ is the regularization coefficient. This formulation defines our prior assumption: all regions in the image should correspond to each text word, while the importance of each word should correspond to its dynamic adjustment based on the text semantics. The OT plan is derived by satisfying these two constraints simultaneously, ensuring the overall consistency of the local alignment.

The entropy-regularized OT problem in (5) is solved using the differentiable Sinkhorn algorithm. In particular, we define the Gibbs kernel as follows:

$$K = \exp\left(-C/\epsilon\right) \tag{6}$$

and iteratively update two scaling vectors u and v

$$u \leftarrow \frac{a}{Kv}, \quad v \leftarrow \frac{b}{K^T u}. \tag{7}$$

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After several Sinkhorn iterations, the OT plan is obtained as follows:

$$\Gamma^* = \mathrm{diag}(u)\, K \mathrm{diag}(v). \tag{8}$$

Since all operations are differentiable, gradients can be back-propagated through the Sinkhorn iterations to optimize the explicit alignment branch.

Each element Γ^* of the transport plan $\Gamma^*_{i,j}$ describes the association strength between the ith image region and the jth text word. Based on this, we construct the alignment loss, which encourages matching image–text pairs to have a higher overall association strength; the calculation formula is as follows:

$$L_{\text{align}} = -\frac{1}{N}\sum_{n=1}^{N} \log\left(\frac{\exp\left(\langle \Gamma_n^*, S_n\rangle_F/\tau_{\text{align}}\right)}{\sum_{m=1,m\neq n}^{N}\exp\left(\langle \Gamma_n^*, S_m\rangle_F/\tau_{\text{align}}\right)}\right) \tag{9}$$

where N denotes the batch size, n is the index of the current positive sample pair, and m represents the indices of the other sample pairs. τ_{align} is the temperature parameter. $\Gamma_n^* \in \mathbb{R}^{P\times L}$ is the OT plan obtained by solving (5). $S_n \in \mathbb{R}^{P\times L}$ is the cosine similarity matrix between image patch features and text token features. The Frobenius inner product $(\Gamma_n^*, S_n)_F = \sum_{i,j}\Gamma_{n,ij}S_{n,ij}$ computes the total weighted similarity under the optimal alignment, serving as a matching score for the positive pair. For negative pairs, we compute $(\Gamma_n^*, S_m)_F$ using the same transport plan Γ_n^* but with similarity matrices S_m from other texts, measuring how well the current image aligns with incorrect texts.

C. Implicit Semantic Guidance Branch

The IB shoulders the responsibility of learning robust global semantic representations, serving as the backbone of the retrieval task. It is not directly supervised by labeled data but instead learns through comparison in the shared semantic space between the image and text. To extract the image and text features $\mathbf{v}_n$ and $\mathbf{t}_n$, the image and text are first passed through the continuous prediction network (MLP), and the corresponding d_h-dimensional vectors are obtained

$$\tilde{\mathbf{v}}_n = \mathrm{MLP}_v(\mathbf{v}_n), \quad \tilde{\mathbf{t}}_n = \mathrm{MLP}_t(\mathbf{t}_n). \tag{10}$$

These vectors are then fine-tuned by optimizing the matching loss function to adjust them for task-specific semantic alignment.

The function for matching in the IB is based on the contrastive learning objective, such as the InfoNCE loss for measuring similarities between image–text pairs.

Where the loss in the I2T direction is similarly expressed as follows:

$$\mathcal{L}_{i2t} = -\frac{1}{N}\sum_{i=1}^{N}\log\left(\frac{\exp\left(s\left(\tilde{\mathbf{v}}_i, \tilde{\mathbf{t}}_i\right)/\tau\right)}{\sum_{j=1}^{N}\exp\left(s\left(\tilde{\mathbf{v}}_i, \tilde{\mathbf{t}}_j\right)/\tau\right)}\right) \tag{11}$$

where the loss in the T2I direction is similarly expressed as follows:

$$\mathcal{L}_{t2i} = -\frac{1}{N}\sum_{i=1}^{N}\log\left(\frac{\exp\left(s\left(\tilde{\mathbf{t}}_i, \tilde{\mathbf{v}}_i\right)/\tau\right)}{\sum_{j=1}^{N}\exp\left(s\left(\tilde{\mathbf{t}}_i, \tilde{\mathbf{v}}_j\right)/\tau\right)}\right). \tag{12}$$

The total contrastive loss is given by

$$\mathcal{L}_{\text{contr}} = \frac{1}{2}\left(\mathcal{L}_{i2t} + \mathcal{L}_{t2i}\right). \tag{13}$$

The key idea of this loss is that it involves a pairwise comparison between image–text pairs, where the similarity function $(\tilde{\mathbf{v}}_i, \tilde{\mathbf{t}}_i)$ measures the similarity $s(\cdot,\cdot)$ between image and text embeddings, and the temperature parameter τ controls the scaling of similarity. The model learns to match the most relevant image–text pairs while minimizing the distance between the representations in the shared embedding space.

D. Adaptive Scalar Quantization Module

In large-scale remote sensing databases, directly using continuous high-dimensional features for cross-modal retrieval not only requires substantial storage space but also incurs high computational complexity for cosine similarity calculations, making it challenging to meet the demands of real-time interactive applications. Binary hash retrieval, by encoding high-dimensional features into compact hash codes and employing efficient Hamming distance computations, can enhance retrieval speed to the millisecond level while significantly reducing storage overhead. Therefore, designing a method capable of converting continuous semantic features into high-quality binary hash codes with minimal or no loss is crucial for enabling large-scale real-time retrieval.

To address this, we design a progressive binarization method. First, the projection head maps the shared space feature $\mathbf{v}_n$ to a K-dimensional space to obtain $\mathbf{h}_n$. During the training phase, we use the hyperbolic tangent function as a continuous relaxation

$$\mathbf{b}_n^{train} = \tanh(\alpha \cdot \mathbf{h}_n) \tag{14}$$

where α is a scaling factor that linearly increases from 1.0 to a larger value (e.g., 10.0) during training. As α increases, the output of tanh gradually approaches ± 1, enabling a smooth transition from continuous features to binary codes.

In the inference stage, we directly obtain the hard binary codes using the sign function

$$\mathbf{b}_n^{\text{test}} = \mathrm{sign}(\mathbf{h}_n). \tag{15}$$

To generate high-quality hash codes, we introduce two optimization objectives.

Quantization Loss: To reduce the gap between continuous relaxation and hard binary codes, we introduce a quantization loss

$$L_{\text{quant}} = \frac{1}{N}\sum_{n=1}^{N}\left\|\mathbf{b}_n^{train} - \mathrm{sg}\left(\mathbf{b}^{\text{test}}\right)\right\|_2^2 \tag{16}$$

where $\mathrm{sg}(\cdot)$ denotes the stop-gradient operator.

The ideal hash codes should have a balanced distribution of $+1$ and -1 at each bit, maximizing the utilization of the Hamming space. We use a variance maximization form to encourage this balance

$$\mathcal{L}_{\text{balance}} = \frac{1}{K}\sum_{k=1}^{K}\left(\frac{1}{N}\sum_{n=1}^{N} b_{\text{train},n,k}\right)^2 \tag{17}$$

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TABLE I
OVERVIEW OF THREE MAINSTREAM CMRSITR DATASETS

	Image			Text
Datasets	Number	Size	Class	Average words
RSICD [19]	10921	224x224	30	23-25
RSITMD [20]	4743	256x256	32	14-16
UCM [21]	2100	256x256	21	19-21

where K denotes the bit index of the hash code.

Through the design outlined above, the DABP module leverages continuous relaxation to approximate binarization during the training process while introducing the quantization loss (L_{quant}) and balance loss (L_{balance}) to ensure that the generated hash codes preserve semantic discriminability and exhibit uniform bit distribution, thereby maximizing the utilization of the Hamming space. During the inference phase, the hard hash codes are directly obtained using the sign function. The storage cost is reduced to only (K) bits per sample (with ($K = 64$) in this article), and similarity computation can be efficiently performed using fast bitwise XOR operations, achieving significant acceleration while maintaining high accuracy. In our implementation, the hash code length K is set to 64. Unless otherwise specified, all EISCG results reported in the main tables of this article are based on 64-bit binary hash codes and Hamming distance retrieval.

E. EISCG Collaborative Training and Optimization

The EISCG framework unifies explicit fine-grained alignment, implicit global representation learning, and efficient hash code generation into a single end-to-end training process through multitask learning. The total loss function of the model is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{contr}} + \lambda_1 \mathcal{L}_{\text{align}} + \lambda_2 \mathcal{L}_{\text{quant}} + \lambda_3 \mathcal{L}_{\text{balance}} \tag{18}$$

where λ_1, λ_2, and λ_3 are hyperparameters used to balance the contributions of different tasks.

IV. EXPERIMENTAL

A. Datasets and Evaluation Metrics

1) Datasets: We conducted our experiments using three widely recognized RSITR datasets: RSICD [19], RSITMD [20], and UCM [21]. These datasets vary in scale, complexity, and the nature of the text descriptions, offering a comprehensive basis for evaluating model performance. RSICD, a commonly used benchmark in this field, consists of 10 921 RS images, each accompanied by five distinct English descriptions, resulting in a total of 54 605 image–text pairs. RSITMD offers more fine-grained and diverse textual descriptions compared to RSICD, focusing on specific objects, spatial relations, and attributes in the images. It contains 4743 remote sensing images, each paired with one textual description, totaling 4743 image–text pairs. UCM is an image-caption dataset containing 2100 high-resolution RS images, each paired with five descriptions, for a total of 10 500 image–text pairs. A summary of the datasets is presented in Table I.

2) Evaluation Metrics: We employed two evaluation metrics, R@K and mR, to assess our model's performance. R@K (which includes R@1, R@5, and R@10) measures the proportion of true values appearing in the top K results. The metric mR represents the average of all R@K values, providing a more thorough evaluation of the model's overall effectiveness.

The other two evaluation metrics are average quantization error (AQE) and balance(bit balance). The AQE is defined as follows:

$$AQE = \frac{1}{N} \sum_{n=1}^{N} \left\| b_n^{\text{train}} - h_n \right\|_2 \tag{19}$$

where b_{train} is the binary code during training and h_n is the continuous feature. The distance between them indicates how much the feature is quantized, with a smaller value suggesting smoother quantization and less information loss.

The balance is defined as follows:

$$Balance = \frac{1}{K} \sum_{k=1}^{K} \left| \frac{1}{N} \sum_{n=1}^{N} b_{n,k}^{\text{test}} \right| \tag{20}$$

where $b_{n,k}^{\text{test}}$ is the kth bit of the test sample's binary code. This value is close to 0, indicating that each bit of the binary code is distributed more evenly, thereby improving space utilization.

Since the proposed DABP module produces binary hash codes for efficient retrieval, we further report commonly used hashing retrieval metrics, including mean Average Precision (mAP) and Average Precision at top K (AP@K). For a given query, AP@K is defined as follows:

$$\text{AP@}K = \frac{1}{\min(K, R)} \sum_{k=1}^{K} P(k) \cdot \delta(k) \tag{21}$$

where R is the number of relevant samples, $P(k)$ denotes the precision at rank k, and $\delta(k) = 1$ if the kth retrieved item is relevant.

The mAP is computed as the mean of AP over all queries

$$\text{mAP} = \frac{1}{Q} \sum_{q=1}^{Q} \text{AP}(q) \tag{22}$$

where Q is the number of queries.

B. Implementation Details

The proposed model is developed within the PyTorch framework and trained using two RTX 3090 GPUs. All input images are uniformly resized to 224 $\times$ 224 pixels, and standard data augmentation strategies, including random horizontal flipping and color jitter, are employed during the training stage. We adopt an end-to-end training paradigm where the parameters of the feature-extraction backbone are kept frozen throughout training, and only the learnable components outside the CLIP backbone are optimized. The optimizer is configured with a learning rate of 1×10^{-4}, weight decay coefficient $\lambda = 0.02$. Due to GPU memory limitations, the batch size is set to 64. For smaller datasets such as RSITMD and UCM, we adopt a gradient accumulation strategy, achieving an effective batch size of 256. All experiments are conducted for 50 training epochs,

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TABLE II
PERFORMANCE COMPARISON WITH STATE-OF-THE-ART METHODS ON THE RSICD DATASET

Method	Vision Backbone	Text Backbone	Image to Text			Text to Image			mR
			R@1	R@5	R@10	R@1	R@5	R@10	
VSE++ [52]	ResNet-18	GRU	3.38	9.51	17.46	2.82	11.32	18.10	10.43
SCAN [53]	Faster R-CNN	BiGRRU	5.85	12.89	19.84	3.71	16.40	26.73	14.23
CAMP [54]	Faster R-CNN	BiGRRU	5.12	12.89	21.12	4.15	15.23	27.81	14.39
AMFMN [20]	ResNet-18	BiGRRU	5.39	15.08	23.40	4.90	18.28	31.44	16.42
HVSA [55]	Faster R-CNN	BiGRRU	7.47	20.62	32.11	5.51	21.13	34.13	16.43
CABIR [56]	ResNet-152	BERT+BiGRU	8.59	16.27	24.13	5.42	20.77	33.58	18.12
IEFT [62]	ViT	BERT	8.78	28.47	43.88	8.38	28.17	44.16	26.97
GALR [57]	ResNet-18	GRU	6.59	19.85	31.04	4.69	19.48	32.13	18.96
SWAN [59]	ResNet-50	GRU	7.41	20.13	30.86	5.56	22.26	37.41	20.61
PIR [60]	ResNet-50	BERT	10.89	26.17	37.79	7.17	25.07	41.06	24.69
CDMAN [61]	PVT	BERT	0.23	24.94	36.83	7.13	25.38	40.35	24.14
MIIA [32]	ViT-B/16	Transformer	11.05	30.17	49.32	13.21	34.74	45.60	30.68
IERR [63]	ViT	Transforme	17.29	35.41	48.58	13.07	35.32	51.31	33.50
GLISA [8]	ViT	Transformer	19.52	40.44	52.28	14.75	39.50	55.46	36.99
RemoteCLIP [58]	ViT-B/32	Transformer	17.02	37.97	51.51	13.71	37.11	54.25	35.26
SWPE [64]	Transformer	Transformer	18.66	39.52	53.61	15.33	40.86	53.73	37.62
WSSCN [65]	Transformer	Transformer	19.58	39.16	51.97	14.77	40.86	56.65	37.17
SGPD [33]	ViT-B/16	Transformer	21.86	43.82	57.09	18.20	43.91	60.14	40.83
EISCG	ViT-B/32	Transformer	21.48	43.32	56.78	18.92	44.28	61.72	41.07
EISCG	ViT-B/16	Transformer	**23.25**	**45.38**	**58.72**	**19.85**	**46.31**	**62.07**	**42.60**

and the model exhibiting the best validation performance is selected for final evaluation. The optimal hyperparameters are determined via grid search on the validation set, with final settings:$\lambda_1 = 0.5$, $\lambda_2 = 0.1$, $\lambda_3 = 0.01$. The temperature for contrastive learning is set to $\tau = 0.05$, and the alignment loss temperature is set to $\tau_{\text{align}} = 0.1$.

C. Comparison With State-of-the-Art Methods

To fairly evaluate the performance of the proposed EISCG framework, we compare it not only with conventional cross-modal retrieval methods but also with approaches based on vision–language pretraining (VLP):

1) *VSE++ [52]:* VSE++ optimizes triplet loss by mining the most challenging negative samples online, enabling more efficient learning of a shared visual-semantic embedding space.
2) *SCAN [53]:* SCAN achieves fine-grained region-word alignment by computing attention between image regions and textual words.
3) *CAMP [54]:* CAMP selects significant information by exploring the interrelationship between text and images.
4) *AMFMN [20]:* AMFMN extracts image features through a multiscale attention mechanism and uses these features to guide text representation.
5) *HVSA [55]:* HVSA enhances the accuracy of text matching by modeling connections between visual features in images.
6) *CABIR [56]:* CABIR employs a cross-attention mechanism to enable information exchange between text and images, primarily focusing on regional-level semantic features.
7) *GALR [57]:* GaLR combines local and global features for matching while enabling them to mutually correct and complement each other.
8) *RemoteCLIP [58]:* RemoteCLIP conducts continual pretraining on remote sensing data based on the CLIP architecture, serving as a foundation model for cross-modal retrieval.
9) *MIIA [32]:* MIIA performs semantic alignment via masked interaction modeling and DCL.
10) *GLISA [8]:* GLISA designs global–local interaction and semantic alignment modules to jointly exploit holistic scene context and local detail information.
11) *SWAN [59]:* SWAN reduces semantic ambiguity by enhancing the fine-grained perception of scenes.
12) *PIR [60]:* PIR enhances the alignment of visual and textual representations through prior instruction-based learning.
13) *CDMAN [61]:* CDMAN improves retrieval accuracy through multigranularity associations and clue-driven feature matching.
14) *IEFT [62]:* IEFT explores the deep connections between visual and textual data, simulating their internal relationships to enhance features.
15) *IERR [63]:* IERR introduces an explicit–implicit reasoning framework that learns local visual–textual relationships, enhancing global image–text matching without extra supervision.
16) *SWPE [64]:* A strong prompt generation module generates fine-grained, global category semantic prompts using an attention mechanism and a pretrained classification model.
17) *WSSCN [65]:* A complete semantic sparse coding network is proposed to construct a comprehensive and reliable feature description.
18) *SGPD [33]:* SGPD converts dense vectors into efficient sparse representations for retrieval via a sparsity-guided partially dense strategy.

Based on the results on three benchmark datasets, as shown in Tables II–IV, we conduct a comprehensive analysis and observe that the proposed EISCG framework consistently outperforms the previous state-of-the-art across all evaluation

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TABLE III
PERFORMANCE COMPARISON WITH STATE-OF-THE-ART METHODS ON THE RSITMD DATASET

Method	Vision Backbone	Text Backbone	Image to Text			Text to Image			mR
			R@1	R@5	R@10	R@1	R@5	R@10	
VSE++ [52]	ResNet-18	GRU	10.38	27.65	39.60	7.79	24.87	36.67	24.83
SCAN [53]	Faster R-CNN	BiGRRU	11.06	25.88	39.38	9.82	29.38	42.12	26.28
CAMP [54]	Faster R-CNN	BiGRRU	9.07	23.01	33.19	5.22	23.32	38.36	22.03
AMFMN [20]	ResNet-18	BiGRRU	10.63	24.78	41.81	11.51	34.69	54.87	29.72
HVSA [55]	Faster R-CNN	BiGRRU	13.20	32.08	45.58	11.43	39.20	57.45	33.16
IEFT [62]	ViT	BERT	15.49	37.61	51.40	11.19	38.09	58.84	35.44
GALR [20]	ResNet-18	GRU	14.82	31.64	42.48	11.15	36.68	51.68	31.41
SWAN [59]	ResNet-50	GRU	13.35	32.15	46.90	11.24	40.40	60.60	34.11
PIR [60]	ResNet-50	BERT	18.36	42.04	55.53	13.36	44.47	61.73	39.25
CDMAN [61]	PVT	BERT	16.46	37.74	51.24	13.82	41.96	59.74	36.83
MIIA [32]	ViT-B/16	Transformer	19.82	52.10	68.11	23.41	45.21	63.29	45.32
IERR [63]	ViT	Transformer	27.66	48.89	61.06	24.43	56.59	72.66	48.55
GLISA [8]	ViT	Transformer	28.21	50.94	62.50	20.50	54.80	72.19	48.41
RemoteCLIP [58]	VIT-B/32	Transformer	27.88	50.66	65.71	22.17	56.46	73.41	49.38
SWPE [64]	Transformer	Transformer	27.88	51.76	64.82	25.27	58.23	75.27	50.54
WSSCN [65]	Transformer	Transformer	28.54	55.31	66.59	24.29	57.43	75.35	51.25
SGPD [33]	ViT-B/16	Transformer	33.21	53.41	70.21	27.03	59.41	79.39	53.78
EISCG	ViT-B/32	Transformer	33.18	56.02	71.78	27.58	62.23	79.42	55.04
EISCG	ViT-B/16	Transformer	**34.85**	**56.37**	**72.15**	**28.75**	**62.30**	**81.45**	**56.13**

TABLE IV
PERFORMANCE COMPARISON WITH STATE-OF-THE-ART METHODS ON THE UCM DATASET

Method	Vision Backbone	Text Backbone	Image to Text			Text to Image			mR
			R@1	R@5	R@10	R@1	R@5	R@10	
VSE++ [52]	ResNet-18	GRU	12.38	44.76	65.71	10.10	31.80	56.85	36.93
SCAN [53]	Faster R-CNN	BiGRRU	12.85	47.14	69.52	12.48	46.86	71.71	43.43
CAMP [54]	Faster R-CNN	BiGRRU	10.95	44.29	65.71	9.90	46.19	76.29	42.22
AMFMN [20]	ResNet-18	BiGRRU	16.67	45.71	68.57	12.86	53.24	79.43	46.08
CDMAN [61]	PVT	BERT	10.95	47.43	73.53	12.23	51.83	85.14	46.90
CABIR [56]	ResNet-152	BERT+BiGRU	15.17	45.71	72.85	12.67	54.19	89.23	48.30
IEFT [62]	ViT	BERT	11.90	47.62	75.71	13.63	53.03	90.45	48.72
MIIA [32]	ViT-B/16	Transformer	18.61	63.41	95.16	21.17	58.36	83.97	56.78
RemoteCLIP [58]	VIT-B/32	Transformer	18.67	61.52	**94.29**	20.48	59.85	83.33	56.36
WSSCN [65]	Transformer	Transformer	24.76	60.48	84.76	21.43	66.57	96.16	59.03
SGPD [33]	ViT-B/16	Transformer	25.71	62.85	88.57	15.61	62.28	93.52	58.09
EISCG	ViT-B/32	Transformer	23.52	66.08	90.47	22.48	66.03	91.02	59.93
EISCG	ViT-B/16	Transformer	**29.24**	**66.50**	91.37	**23.15**	**67.09**	**96.80**	**62.36**

metrics. In particular, on the RSICD dataset in Table II, EISCG achieves a pronounced gain in R@1, exceeding SGPD (21.86%) by 1.89%. Moreover, EISCG delivers the best recall performance for both I2T and T2I retrieval, with the mean recall (mR) markedly surpassing all competing methods. On the RSITMD dataset in Table III, EISCG again attains the highest R@1 (34.85%) in the I2T direction while also achieving the strongest R@10 in the T2I direction. The resulting mR reaches 56.13%, outperforming WSSCN (51.25%) and SGPD (53.78%) by 4.88% and 2.35%, respectively. On the UCM dataset in Table IV, EISCG shows a comprehensive advantage across multiple metrics, particularly in T2I retrieval where R@5 (67.09%) and R@10 (96.80%) are substantially higher than those of all compared methods. EISCG further achieves the best mR of 62.36%, exceeding SGPD (58.09%) and WSSCN (59.03%) by 4.27% and 3.33%, respectively. It is worth noting that the EISCG results reported in Tables II–IV are based on 64-bit binary hash codes, whereas all comparison methods perform retrieval using continuous features. Despite employing a more compact representation, EISCG consistently achieves higher retrieval accuracy than existing methods, which further demonstrates the effectiveness of the proposed explicit–implicit collaborative guidance mechanism.

Meanwhile, since the final retrieval can be performed using compact binary hash codes and Hamming distance, we compare our method with five state-of-the-art deep hashing-based cross-modal retrieval methods on the RSICD dataset in terms of mAP@20, including CPAH [8], DJSRH [10], JDSH [11], as well as DMFH and CACH [12]. We also provide commonly used bit-length comparisons, including 16-, 32-, and 64-bit settings. As shown in Table V, as the hash code length increases from 16 to 64 bits, the mAP@20 of all methods consistently improves. EISCG significantly outperforms existing methods across all code lengths and in both retrieval directions. In particular, at 64 bits, EISCG achieves 0.881 and 0.896 for I2T and T2I retrieval, respectively, surpassing the previous best method CACH (0.867/0.882) by 0.014 in both cases. At 16 bits, EISCG further demonstrates a notable advantage under short-code settings, achieving 0.784 (I2T) and 0.828 (T2I), which are higher than CACH by 0.027 and 0.015, respectively.

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TABLE V
COMPARISON OF MAP@20 RESULTS ON THE RSICD DATASET WITH STATE-OF-THE-ART METHODS

Methods	Image to Text			Text to Image		
	16 bits	32bits	64bits	6 bits	32bits	64bits
CPAH [66]	0.428	0.587	0.636	0.452	0.598	0.667
DJSRH [67]	0.411	0.665	0.688	0.422	0.685	0.705
JDSH [68]	0.385	0.720	0.796	0.418	0.751	0.799
DUCH [69]	0.684	0.791	0.836	0.697	0.780	0.824
DMFH [70]	0.703	0.746	0.841	0.712	0.793	0.844
CACH [71]	0.757	0.836	0.867	0.813	0.844	0.882
EISCG	**0.784**	**0.852**	**0.881**	**0.828**	**0.859**	**0.896**

TABLE VI
ABLATION STUDY ON EACH COMPONENT OF EISCG ON RSICD AND RSITMD DATASETS

Methods	Compoments			Image → Text			Text → Image			mR
	EB	CT	DABP	R@1	R@5	R@10	R@1	R@5	R@10	
RSICD Dataset										
Baseline				18.50	38.75	51.80	15.20	37.10	52.80	35.69
+EB	✓			20.15	40.23	53.91	17.85	39.42	55.61	37.86
+CT		✓		21.33	42.45	55.61	19.10	41.36	57.20	39.51
+ DABP			✓	19.10	40.85	53.92	16.75	39.90	55.80	37.72
+EB+CT	✓	✓		22.85	44.20	57.35	20.40	43.15	59.10	41.18
+EB+DABP	✓		✓	21.95	42.80	56.15	19.55	42.05	58.25	40.13
EISCG	✓	✓	✓	**23.25**	**45.38**	**58.72**	**20.65**	**44.27**	**60.39**	**42.11**
RSITMD Dataset										
Baseline				24.73	46.82	63.15	21.89	44.27	66.84	44.62
+EB	✓			27.16	49.38	67.42	24.81	47.63	70.35	47.79
+CT		✓		29.85	51.74	69.08	26.53	49.92	72.61	50.29
+ DABP			✓	25.94	48.57	65.33	23.42	46.88	68.47	46.43
+EB+CT	✓	✓		31.47	53.86	71.59	28.05	52.14	74.82	52.15
+EB+DABP	✓		✓	28.31	50.95	68.74	25.97	49.21	71.53	49.12
EISCG	✓	✓	✓	**34.85**	**56.37**	**72.15**	**28.75**	**62.30**	**81.45**	**56.13**

D. Ablation Experiments

To evaluate the contribution of each component to the EISCG framework, we conducted systematic ablation experiments. First, we analyzed the effect of separate elements in the EISCG structure by incorporating certain components. After that, a thorough study was carried out to establish the exact function of the four loss functions in the suggested approach. Then, we identified the optimal configuration of the total loss function. Finally, experiments on the mask ratio were set about the context-aware TF-IDF strategy.

1) EISCG Module: We designed several model variants with different configurations to evaluate the contribution of the EB, IB, context-aware TF-IDF (CT) masking strategy, and DABP module to the overall performance. The Baseline model uses only the IB, and EISCG represents the full model. We conducted detailed experimental analysis on two public datasets (RSICD and RSITMD), as shown in Table VI. On the RSICD dataset, after introducing EB, the mR improved by 2.17%, indicating that incorporating the EB as the retrieval backbone significantly enhances the performance of traditional implicit methods, validating the necessity of explicit–implicit collaboration. This is primarily attributed to the fine-grained alignment capability introduced by the OT module, which demonstrates the effectiveness of OT in capturing local semantic correspondences. After adding CT, R@1 increased to 21.33%, demonstrating that dynamically adjusting word weights based on semantic context provides more precise fine-grained supervision signals. With the introduction of DABP, although the accuracy improvement was limited, the ABP endowed the model with millisecond-level hash retrieval capability, achieving a balance between accuracy and efficiency. When all three components (EB, CT, and DABP) were combined, the model achieved the best balance between performance and efficiency. On the RSICD dataset, the I2T R@1 reached 23.25%, and the T2I R@1 reached 20.65%. On the RSITMD dataset, the I2T R@1 reached 34.85%, and the T2I R@1 reached 28.75%. These results demonstrate that the full model outperforms any component combination, validating the necessity and effectiveness of the collaborative operation of all components.

To further validate the contribution of the DABP module to retrieval efficiency, we measured the retrieval time, QPS, and storage overhead for different configurations on the RSICD dataset, and compared the results with several representative SOTA methods. As shown in Table VII, after incorporating the DABP module, the retrieval efficiency of EISCG significantly improves: the retrieval time drops from 3.0 to 0.9 ms, with QPS increasing from 333 to 1042, achieving a 3.1× speedup. The storage requirement decreases from 2.83 to 0.008 GB, resulting in a 353× reduction in large-scale storage overhead. Meanwhile, the retrieval accuracy is not compromised by quantization; the mR slightly increases from 42.12% to 42.60%, achieving a favorable balance between efficiency and accuracy. Compared to other SOTA methods, EISCG (with DABP) achieves faster retrieval (0.9 ms) while significantly outperforming SGPD (3.5 ms), MIIA (3.2 ms), and RemoteCLIP (2.8 ms) by 2–3 times. These results demonstrate that the DABP module effectively preserves accuracy while enabling significant retrieval improvements, making it particularly suitable for large-scale real-time remote sensing applications.

To comprehensively evaluate the computational efficiency of the EISCG framework, we compared the training time, inference time, parameter count, FLOPs, and retrieval accuracy (mR) of EISCG (with/without OT module) against several representative SOTA methods on the RSICD dataset. The results are shown in Table VIII. Comparing EISCG (w/o OT) and EISCG (full), the introduction of the OT module increases the training time by 42.4%, inference time by 34.8%, and parameter count and FLOPs by approximately 6%. In exchange, mR improves from 36.35% to 42.60%, with a gain of 6.25 percentage points. This indicates that the computational overhead introduced by the OT module is justified, as the accuracy improvement is significant, and the inference time remains within an acceptable range. Compared to SOTA methods, EISCG (full) has significantly lower parameter count (12.8 M) and FLOPs (9.2 G) than all compared Transformer-based methods. The inference time (3.1 ms) is comparable to MIIA and SGPD, and better than GLISA. The training time (178 s/epoch) is slightly higher than MIIA and RemoteCLIP, but lower than SGPD. Considering the superior accuracy, EISCG demonstrates competitive overall computational efficiency among SOTA methods. In conclusion, although the OT module introduces some computational overhead, the

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TABLE VII
EFFICIENCY ANALYSIS OF THE DABP MODULE ON THE RSICD DATASET

Method	Feature Type	Retrieval Time (ms/query) ↓	QPS ↑	mR (%) ↑	Storage(GB 1M samples)
MIIA	Continuous (float32)	3.2	312	30.68	2.86
RemoteCLIP	Continuous (float32)	2.8	357	35.26	2.69
SGPD	Continuous (float32)	3.5	286	40.83	3.81
EISCG (w/o DABP)	Continuous (float32)	3.0	333	42.12	2.83
EISCG (w/ DABP)	**Binary (64-bit)**	**0.9**	**1042**	**42.60**	**0.008**

TABLE VIII
EFFICIENCY ANALYSIS OF THE OT MODULE ON THE RSICD DATASET

Method	Training Time (s/epoch) ↓	Inference Time (ms/query) ↓	Params (M) ↓	FLOPs (G) ↓	mR (%) ↑
AMFMN	85	1.8	18.5	12.3	16.42
MIIA	156	3.2	28.4	18.6	30.68
GLISA	168	3.4	30.1	20.3	36.99
RemoteCLIP	142	2.8	26.5	16.2	35.26
SGPD	185	3.5	32.7	21.5	40.83
EISCG (w/o OT)	**125**	**2.3**	**12.1**	**8.7**	**36.35**
EISCG (full)	**178**	**3.1**	**12.8**	**9.2**	**42.60**

TABLE IX
ABLATION STUDY OF ALL LOSS FUNCTIONS OF EISCG ON RSICD DATASET

Losses				Image to Text			Text to Image			mR	AQE (↓)	Balance(↓)
L_{contr}	L_{align}	L_{quant}	L_{balance}	R@1	R@5	R@10	R@1	R@5	R@10			
✓				18.50	38.75	51.80	15.20	36.10	57.75	36.35	N/A	N/A
✓	✓			22.01	43.87	56.93	19.85	41.36	61.60	40.94	N/A	N/A
✓	✓	✓		23.05	45.10	58.20	20.65	44.27	61.43	42.12	0.16	0.21
✓	✓		✓	22.91	44.90	57.88	20.50	44.10	61.09	41.90	N/A	N/A
✓		✓	✓	19.84	40.12	53.06	17.67	38.12	57.23	37.67	0.18	0.05
✓	✓	✓	✓	**23.25**	**45.38**	**58.72**	**20.65**	**45.00**	**62.60**	**42.60**	**0.15**	**0.03**

TABLE X
ABLATION STUDY OF WEIGHTS λ_1, λ_2, AND λ_3 IN THE TOTAL LOSS FUNCTION ON RSICD DATASET

λ_1	R@1	mR	AQE(↓)	Balance(↓)	λ_2	R@1	mR	AQE(↓)	Balance(↓)	λ_3	R@1	mR	AQE(↓)	Balance(↓)
0.0	19.84	37.67	0.18	0.05	0.0	22.91	41.90	0.42	0.04	0.0	23.05	42.12	0.16	0.21
0.2	21.05	39.52	0.16	0.04	0.05	23.05	42.05	0.25	0.04	0.005	23.15	42.35	0.15	0.10
0.5	**23.25**	**42.60**	**0.15**	**0.03**	**0.1**	**23.25**	**42.60**	**0.15**	**0.03**	**0.01**	**23.25**	**42.60**	**0.15**	**0.03**
0.8	22.91	41.91	0.15	0.03	0.2	23.10	42.40	0.12	0.03	0.02	23.20	42.45	0.15	0.02
1.0	22.03	40.89	0.16	0.04	0.5	22.80	41.85	0.09	0.03	0.05	23.10	42.25	0.16	0.01
1.5	21.25	39.12	0.17	0.04	0.75	22.35	41.10	0.08	0.03	0.075	23.05	42.18	0.16	0.01
2.0	20.56	38.78	0.18	0.05	1.0	22.01	40.12	0.08	0.03	0.1	23.00	42.10	0.16	0.01

significant accuracy gain it provides justifies its inclusion. Moreover, EISCG (full) remains competitive in terms of parameter count, FLOPs, and inference speed compared to SOTA methods.

2) Loss Function: To validate the necessity of each component in the total loss function $L_{\text{total}} = L_{\text{contr}} + \lambda_1 L_{\text{align}} + \lambda_2 L_{\text{quant}} + \lambda_3 L_{\text{balance}}$, we conducted a systematic ablation experiment on the RSICD dataset, as shown in Table IX. The retrieval accuracy (mR) and hash code quality (measured by AQE and balance) were also evaluated.

After introducing the explicit fine-grained alignment loss L_{align}, the R@1 on I2T improved by 3.51 percentage points (from 18.50% to 22.01%), and the R@1 on T2I increased by 4.65% to 19.85%. The mR significantly improved by 4.59%, reaching 40.94%, demonstrating a clear improvement. This highlights the necessity of introducing explicit alignment loss to collaborate with the implicit loss function. With the addition of L_{quant} (third row) on top of the existing L_{contr} and L_{align}, the mR further increased to 42.12% (+1.18%), and the AQE was 0.16. This suggests that the quantization constraint played a regularizing role during training, making the continuous feature representation more compact and contributing to accuracy improvement. Adding L_{balance} (fourth row) alone resulted in a smaller mR improvement (+0.96%) compared to adding L_{quant}. Finally, in the last row (full loss), all metrics reached their optimal values: the highest retrieval accuracy (mR = 42.60%), the best hash code quality (AQE = 0.15, the lowest, and Balance = 0.03, closest to 0). This confirms the effectiveness of the collaboration of the four loss functions: L_{contr} and L_{align} strengthen the semantic alignment foundation, while L_{quant} and L_{balance} jointly optimize the information fidelity and coding efficiency of the hash codes, ensuring accuracy.

3) Weight Factor: To further explore the loss function $L_{\text{total}} = L_{\text{contr}} + \lambda_1 L_{\text{align}} + \lambda_2 L_{\text{quant}} + \lambda_3 L_{\text{balance}}$, we investigate

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TABLE XI
JOINT GRID SEARCH RESULTS OF THE WEIGHTS (SOME REPRESENTATIVE COMBINATIONS)

$(\lambda_1, \lambda_2, \lambda_3)$	mR	Relative default increase
(0.5, 0.1, 0.01) - Default	42.60	-
(0.6, 0.1, 0.01)	42.58	-0.02
(0.5, 0.15, 0.01)	42.65	+0.05
(0.5, 0.1, 0.02)	42.55	-0.05
(0.4, 0.12, 0.015)	42.61	+0.01

the effects of each weight component $(\lambda_1, \lambda_2, \lambda_3)$ by tuning their values and validating the model's performance on the RSICD dataset through network retrieval performance experiments, as shown in Table X. We first use default values for the first two weights and adjust the third weight. The default values are: $\lambda_1 = 0.5, \lambda_2 = 0.1, \lambda_3 = 0.01$.

Query-Based Retrieval Experiment: Set $\lambda_2 = 0.1, \lambda_3 = 0.01$, and test $\lambda_1 \in \{0.0, 0.2, 0.5, 1.0, 10.0\}$.

Weight Adjustment Experiment: Set $\lambda_1 = 0.5, \lambda_3 = 0.01$, and test $\lambda_2 \in \{0.0, 0.05, 0.1, 0.2, 0.5, 1.0\}$.

Balance Adjustment Experiment: Set $\lambda_1 = 0.5, \lambda_2 = 0.1$, and test $\lambda_3 \in \{0.0, 0.005, 0.01, 0.02, 0.05, 0.1\}$.

Combined Network Retrieval: Test on combined queries within the local dataset to verify whether the integrated combinations produce improved results, as shown in Table X.

As shown in Table X, experiments indicate that the model performs well within the λ_1 range of 0.2–0.8, with optimal performance achieved at $\lambda_1 = 0.5$. When $\lambda_1 = 0$, performance significantly declines (−4.93 points), reaffirming the necessity of explicit supervision. Beyond $\lambda_1 > 1.0$, performance begins to deteriorate, suggesting that overly fine-grained alignment interferes with global feature learning. λ_2 primarily influences AQE. Increasing λ_2 significantly reduces AQE, but retrieval accuracy first increases then decreases. The optimal balance between accuracy and AQE is achieved at $\lambda_2 = 0.1$. λ_3 primarily controls hash code balance. Increasing λ_3 from 0 to 0.01 substantially improves balance (from 0.21 to 0.03), while accuracy remains largely unaffected or even slightly improves. Beyond $\lambda_3 > 0.01$, balance improvement becomes marginal and accuracy begins to decline slightly. This indicates that a slight balancing constraint ($\lambda_3 = 0.01$) is sufficient to effectively regulate the hash code distribution, while overly stringent constraints prove counterproductive. As shown in Table XI, joint grid search reveals a flat optimal region near the default weights (0.5, 0.1, 0.01), where performance differences between adjacent combinations are negligible (<0.1 points).

4) Mask Ratio: As shown in Fig. 3, experiments were conducted on the RSICD and RSITMD datasets. We used mR as the final test metric. The bar chart clearly shows that introducing any proportion of masking ($\rho > 0$) significantly outperforms the baseline without masking ($\rho = 0.0$). On the RSICD dataset, the best mask improves the baseline by 6.80 points, and on the RSITMD dataset, it improves by 9.70 points. This strongly demonstrates that context-aware masking is an effective data augmentation and regularization technique. The image clearly shows that model performance increases and then decreases in a parabolic trend as ρ increases. For both

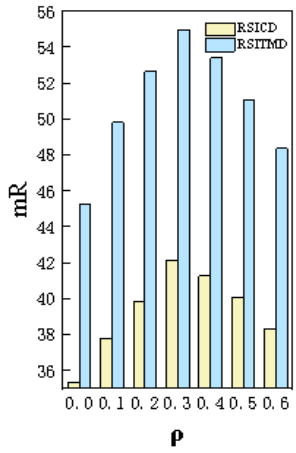


Fig. 3. Analysis of ablation experiment results on the context masking ratio on the RSICD and RSITMD datasets.

RSICD and RSITMD, $\rho = 0.3$ is the optimal value. When ρ exceeds 0.4, the performance on both datasets declines steadily. This indicates that when too much critical semantic information is masked ($\rho > 40\%$), reconstructing the correct cross-modal alignment becomes excessively difficult, which may cause the model to learn noisy or unstable associations, ultimately degrading the quality of the final representation. Therefore, we set the text mask ratio to 0.3 to pursue the best performance of the model for retrieval.

E. Qualitative Results

Fig. 4 illustrates the visual attention maps of our proposed EISCG, highlighting its responses to individual words within the input text. The heatmaps shown in Fig. 4 are generated in an offline analysis manner for interpretability purposes. In particular, we recompute the OT plan Γ^* using the learned feature representations after training, in order to visualize the learned word-region alignment patterns. Therefore, these visualizations do not reflect the inference pipeline and do not introduce any additional computational overhead during deployment. It can be clearly observed that EISCG is capable of accurately identifying different objects within the same image. For geographic environmental elements such as rivers, ponds, and trees, the attention is precisely focused on the corresponding natural regions in the image, indicating that the model possesses strong perceptual ability with respect to scene-level geographic semantics. For man-made objects, such as buildings, vehicles, and roads, the model is able to effectively distinguish them and assign distinct attention patterns, even when these objects are spatially adjacent or interleaved. However, when multiple objects overlap or occlude each other, the model exhibits difficulty in consistently maintaining independent attention for each object, which may limit its effectiveness in densely populated target scenes.

Fig. 5 presents qualitative results of our proposed EISCG on the RSITR task, along with a comparison to the results of the best-performing SGPD model. Given a query, the top five retrieved results are arranged from left to right. Correct matches are highlighted in green, while incorrect

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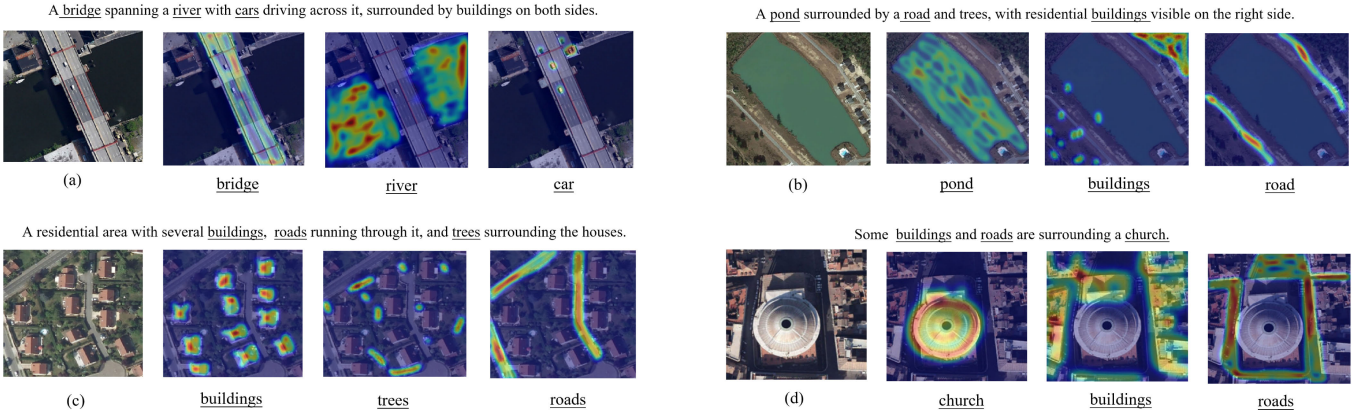


Fig. 4. Visualizations of attention-based heatmaps on the RSICD dataset. (a)–(d) Image attention maps with respect to different words.

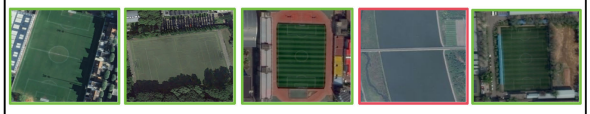
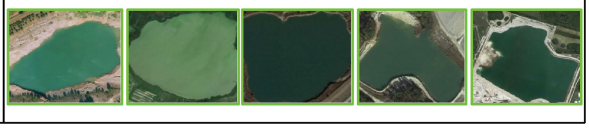
Query		Top-5 Retrival Result	Query		Top-5 Retrival Result
	SGPD	The planes are accelerating for takeoff on the runway. The airport is surrounded by a thick layer of snow. A train is crossing the airport apron. there are many planes with same size in the airport . The planes are flying in formation above the airport.	many buildings and some green trees are around a playground near a road in a school .	SGPD	
	EISCG	many planes are parked next to a long building in an airport . there are many planes with same size in the airport . Some aircrafts are near industrial buildings. some planes are parked near an airport with parking lot . there are two long narrow shapes on the bareland .		EICSG	
	SGPD	there are a lot of bare land near the airport . the bareland with red stripes has a white fog like shadows . many green trees are in two sides of a river with a bridge over it . the spring pool is surrounded by lush trees and houses . the picture shows a dual-layer bridge with vehicles passing by.	there is a turquoise-colored lake surrounded by rocky terrain.	SGPD	
	EISCG	there are many running car on the bridge. many green trees are in two sides of a river with a bridge over it . there are three baseball fields on the land . the image depicts a bridge spanning across a river. the picture shows a dual-layer bridge with vehicles passing by.		EISCG	

Fig. 5. Given a query image or text from the RSICD test set, the top five retrieval results for both SGPD and our model EISCG are listed for comparison. Correct retrieval results are marked in green, while incorrect results are marked in red.

ones are marked in red. It can be observed that even for small objects such as vehicles and aircraft, our method is able to retrieve accurate matches in most cases. However, in certain situations, some results labeled as "incorrect" may still correspond to semantically reasonable captions. For example, the third retrieved result in the first query accurately describes the image content despite being marked as incorrect. This discrepancy may stem from the fact that dataset annotations are often short and lack fine-grained semantic details, leading to potential ambiguities during evaluation. Consequently, the semantic complexity and annotation quality of the dataset may influence the performance of EISCG. Addressing this issue remains an important direction for future work.

V. Conclusion

This study proposes an EISCG framework for remote sensing image–text retrieval, providing an innovative solution that jointly considers retrieval accuracy, computational efficiency, and interpretability. By leveraging the collaboration between explicit and implicit semantic branches, the proposed framework effectively alleviates the semantic gap and the loss of fine-grained information commonly encountered in conventional cross-modal retrieval methods. In addition, an adaptive binary hashing strategy is introduced to enable efficient large-scale retrieval over remote sensing datasets. Extensive experimental results demonstrate that EISCG achieves significant performance improvements across multiple public benchmarks while exhibiting strong generalization capability and practical applicability. Nevertheless, the limited semantic complexity of existing datasets constrains the training and inference capacity of the model. In future work, we will further investigate the adaptability of the proposed framework to more complex scenes and multilingual retrieval scenarios.

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